

Serverless Architectures TASK – 2

1. Create an S3 bucket named 'music-uploads-demo' and upload a sample MP3 file to it using the AWS Management Console.

Ans :

1. Sign in to the AWS Management Console:

<https://console.aws.amazon.com/>

2. In the AWS search bar, search for "S3" and open the Amazon S3 service.

3. Click "Create bucket".

4. Enter this bucket name:

music-uploads-demo

5. Choose an AWS Region close to you.

6. Leave the other settings at their defaults, including "Block all public access".

7. Scroll down and click "Create bucket".

8. Open the newly created "music-uploads-demo" bucket.

9. Click "Upload".

10. Click "Add files".

11. Select a sample MP3 file from your computer, for example:
sample-song.mp3

12. Click "Upload".

13. Wait for the upload to complete.

14. Return to the bucket's "Objects" tab.

15. Verify that your MP3 file appears in the list.

16. The S3 bucket is now ready to store your music uploads.

2. Write an AWS Lambda function in Python called `log_upload_event` that logs the filename and upload timestamp whenever a new file is added to your 'music-uploads-demo' S3 bucket.

Ans :

```
import json
```

```
import logging
```

```
from urllib.parse import unquote_plus
```

```

logger = logging.getLogger()
logger.setLevel(logging.INFO)

def log_upload_event(event, context):
    for record in event.get("Records", []):
        filename = unquote_plus(record["s3"]["object"]["key"])
        upload_timestamp = record.get("eventTime", "Unknown")

        logger.info(
            "File uploaded: %s | Upload timestamp: %s",
            filename,
            upload_timestamp
        )

    return {
        "statusCode": 200,
        "body": json.dumps("Upload event logged successfully")
    }

```

3. Configure an event trigger so that your `log_upload_event` Lambda function is automatically invoked every time a new file is uploaded to the 'music-uploads-demo' S3 bucket.

Ans :

1. Sign in to the AWS Management Console:

<https://console.aws.amazon.com/>

2. Open "S3" from the AWS services search bar.
3. Open the bucket:
music-uploads-demo
4. Go to the "Properties" tab.
5. Scroll down to "Event notifications".
6. Click "Create event notification".
7. Enter an event name, for example:
log-upload-event
8. Under "Event types", select:
Object Creation → All object create events
9. Leave the prefix and suffix filters empty so every uploaded file triggers the function.
10. Under "Destination", select:
Lambda function

11. Select the Lambda function:

`log_upload_event`

12. Click "Save changes".

13. If AWS asks for permission to allow S3 to invoke the Lambda function, approve the permission.

14. Upload a new sample MP3 file to:

`music-uploads-demo`

15. Open AWS Lambda → `log_upload_event` → Monitor → View CloudWatch logs.

16. Open the latest log stream.

17. Verify that you see a message similar to:

File uploaded: sample-song.mp3 | Upload timestamp: 2026-09-30T12:30:00.000Z

18. This confirms that S3 automatically invokes `log_upload_event` whenever a new file is uploaded.

4. Simulate uploading a new file to your S3 bucket and verify that your Lambda function runs by checking the logs in AWS CloudWatch.
Hint: Look for the

Lambda function's log group in CloudWatch to see the output for each upload event.

Ans :

1. Sign in to the AWS Management Console:

<https://console.aws.amazon.com/>

2. Open "S3" and select the bucket:

music-uploads-demo

3. Click the "Upload" button.

4. Click "Add files" and select a sample file, for example:

test-upload.mp3

5. Click "Upload" and wait until the upload completes.

6. Open the AWS Lambda service.

7. Select the Lambda function:

log_upload_event

8. Open the "Monitor" tab.

9. Click "View CloudWatch logs".

10. Select the latest log stream under the Lambda function's log group:

`/aws/lambda/log_upload_event`

11. Look for a log entry similar to:

File uploaded: test-upload.mp3 | Upload timestamp: 2026-09-30T12:30:00.000Z

12. You should also see START and END/REPORT entries showing that the Lambda invocation occurred.

13. If the filename and timestamp appear in the CloudWatch logs, the S3 → Lambda event trigger is working successfully.

5. Use ChatGPT or GitHub Copilot to suggest an additional event-driven use case for S3 and Lambda (other than logging uploads), and describe in 2-3 lines how you would implement it.

Ans : When an MP3 is uploaded to the music-uploads-demo S3 bucket, an S3 Object Created event can trigger a Lambda function automatically.

The Lambda function could validate the audio file, extract metadata such as duration/title, and store the results in DynamoDB or another S3 object.